

RESEARCH ARTICLE

How Well Does a Pointwise Mutual Information Threshold Identify Questionable Wikidata Biomedical Relations?

An independent-verdict evaluation of 40,725 flagged relations

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ABSTRACT

Background. Turki et al. (2024) flagged Wikidata biomedical relations whose entities show weak or no association in PubMed, using pointwise mutual information (PMI) between MeSH keywords. Relations with PMI below 2, or with no co-occurrence (PMI = -10), were sent for human validation. The threshold was taken from the shape of the PMI distribution, not tested against whether the relations are correct.

Objective. To measure how often the flagged relations are judged accurate by an independent automated judge, and whether PMI predicts that judgement.

Methods. We analysed the 15,925 relations with PMI < 2 and the 24,800 relations with no PubMed association. Each had received one verdict (true, maybe or false) from a TypeSafe SDK call (Jev) that saw only the three English labels of the triple. We computed the share of true verdicts per PMI value with Wilson intervals, fitted logistic regressions with property fixed effects and subject-clustered errors, computed within-property AUCs with stratified bootstrap intervals, and compared the no-association group with the weak-association group.

Results. Jev judged 65.3% (95% CI 64.6–66.1) of relations with PMI < 2 and 69.9% (69.3–70.4) of relations with no PubMed association to be accurate. Pooled over relation types, the accuracy rate is almost flat above PMI -1, because non-biomedical relations such as diplomatic relations are judged accurate far more often and are concentrated at low PMI. Within relation type, each additional PMI unit raised the odds of a true verdict by a factor of 1.37 (1.32–1.42), against 1.08 unadjusted, and negative PMI halved the odds relative to PMI 0–2 (OR 0.50). Discrimination was modest (within-property AUC 0.60). Relations with no PubMed association were judged accurate more often than weakly associated ones (69.2% vs 56.6% among biomedical properties; adjusted OR 1.61).

Conclusions. Under this judge, most flagged relations are not wrong, so the flag is better used as a review queue than as an error list. PMI carries a real but modest signal within relation types. The supplied data contain no relations above PMI 2 and therefore cannot show whether 2 is a meaningful boundary. We provide a ready-to-run test for that question and the sample size it needs.

Keywords: Wikidata; MeSH; PubMed; pointwise mutual information; knowledge graph quality; relation verification; threshold selection

1. Introduction

Wikidata is a collaboratively edited knowledge graph and a large source of structured biomedical information (Vrandečić, 2013; Turki et al., 2019). Its statements come from bulk imports, bots and manual edits, so quality is uneven, and auditing it against independent evidence is a practical need.

Turki et al. (2024) proposed one such audit. They aligned Wikidata items with MeSH descriptors, computed PMI between MeSH keywords from PubMed counts, and flagged Wikidata relations whose entity pairs are weakly or not at all associated in PubMed. They reported 40,725 flagged relations: 15,925 with PMI below 2 and 24,800 with no PubMed association.

Two questions remain open. First, **how many flagged relations are actually problematic?** A PMI flag measures co-indexing in PubMed; it does not measure whether a relation is true. Second, **does**

PMI track plausibility at all, and is 2 a sensible cut-off? The threshold was derived from the PMI distribution and from benchmarks on word-pair relatedness, not from the correctness of knowledge-graph statements.

We address these questions with an independent verdict for every flagged relation. The claim is deliberately narrow: the verdict comes from an automated service that sees three labels, so it measures *plausibility as judged by that service*, not verified truth. Our contributions are (i) a quantitative description of how that plausibility varies with PMI; (ii) a demonstration that pooling relation types hides most of the PMI signal; and (iii) a test design, with code, for the question these data cannot answer.

Research questions.

1. **RQ1.** What fraction of relations with $\text{PMI} < 2$ is judged accurate?
2. **RQ2.** What fraction of relations with no PubMed association is judged accurate?
3. **RQ3.** How does the rate vary with PMI and with relation type?
4. **RQ4.** Does PMI predict the verdict after controlling for relation type?
5. **RQ5.** Are relations with no co-indexing judged differently from weakly co-indexed ones?

2. Background

2.1 PMI between MeSH keywords

For two MeSH keywords x and y , Turki et al. (2024; their Eq. 1, following Church and Hanks, 1990, and Fano, 1961) define

$$\begin{aligned} \text{PMI}(x, y) &= \log_2 (N(x, y) \cdot R / [N(x) \cdot N(y)]) \quad \text{if } N(x, y) > 0 \\ \text{PMI}(x, y) &= -10 \quad \text{if } N(x, y) = 0 \end{aligned} \quad (1)$$

where R is the number of PubMed publications, $N(x)$ and $N(y)$ are the numbers of search results for each keyword, and $N(x, y)$ is the number for the pair. A PMI of 2 means the pair co-occurs four times as often as independence predicts; a negative PMI means less often than expected. The value -10 is a default for pairs that never co-occur and lies outside the range of computed values (-5.73 to 23.10 in the original study). PMI is known to over-weight rare events and to be unstable at low counts (Damani, 2013), which matters here because many flagged relations involve rare entities.

2.2 How the threshold was chosen

Turki et al. extract Wikidata relations between items that have MeSH identifiers, round their PMI down to integers, and take the **mode** of the rounded values as the threshold. They report a peak at 2.0 and cite earlier work that used PMI thresholds for word-pair relatedness (Han et al., 2012; Kennedy and Hirst, 2012). Relations below the threshold are “flagged for human validation” (their Section 3.2) and elsewhere described as “not considered valid” (Section 4.2). The threshold therefore reflects the shape of a distribution, not a measured change in correctness.

2.3 The flagged sets

The original study reports 96,341 relations between items with accurate MeSH identifiers, of which 15,925 (16.5%) fall below PMI 2. A further 24,800 relations (22.7% of the 109,302 relations about MeSH-linked entities) have no PubMed association. The supplied verdict files contain exactly these two sets; their per-property counts match the original Table 8.

3. Data

Three layers of data are involved and are kept separate throughout (Table 1).

Table 1. Data layers.

Layer	Content	Source
Source	Wikidata statements between items with a MeSH descriptor ID (P486)	Turki et al. (2024), queries L1 and L2
Derived	PMI per relation (Eq. 1), including the -10 default	Turki et al. (2024); precomputed in the input files
Outcome	One verdict per relation: true, maybe or false	TypeSafe SDK call (Jev) in the original notebook

Two files were analysed. The first holds the 15,925 relations with $PMI < 2$ (minimum -5.73 , maximum 1.9998). The second holds the 24,800 relations with $PMI = -10$. Each row has the Wikidata subject, property and object (IDs and English labels), the MeSH IDs, the PMI and the verdict. English labels had been fetched from Wikidata and written into the files by the original notebook. Table 2 summarises the files.

Table 2. Composition of the two verdict files.

Characteristic	PMI < 2	No PubMed association
Relations	15,925	24,800
Distinct properties	129	118
Distinct subjects / objects	5,853 / 1,876	10,340 / 4,152
Subjects with a MeSH descriptor ID (D...)	15,925	7,062
Subjects with a supplementary-concept ID (C...)	0	17,732 (71.5%)
Duplicate (subject, property, object) rows	46	146
Subject / object label left as a bare QID	33 / 46	36 / 157
Rows with identical MeSH ID at both ends	0	44

Twenty-six (subject, property, object) triples occur in both files; the files do not show why.

The two groups differ in composition, not only in PMI. The no-association group is dominated by MeSH supplementary concepts (mostly specific chemicals), which are indexed in few articles.

4. Methods

4.1 The outcome: a Jev verdict

Each relation was judged once by a call to the TypeSafe SDK (*system_one*). The system text states that the goal is to decide whether the triple “represents a valid and meaningful relationship according to established scientific knowledge and terminology”, and the instruction names the subject, predicate and object by their English labels. The answer had to be one of true, maybe or false; no definitions of these categories were supplied. No abstracts, MeSH data, PubMed identifiers, Wikidata descriptions or ontology context were passed, and the call returns no explanation or confidence. How the service reaches its verdict is not documented in the available materials, and whether it is deterministic is unknown.

We therefore describe a true verdict as “judged accurate by Jev”. It is not literature verification and was not checked by humans.

4.2 Rate per PMI value

PMI is rounded down to an integer, as in the original study; bin b covers $[b, b+1)$. For each bin we report the share of true verdicts with a 95% Wilson score interval (Wilson, 1927). Half-unit bins are available in the notebook output. The no-association group has the single value -10 .

4.3 Non-biomedical relations

The original study notes that relations such as diplomatic relation and shares border with are poorly covered by PubMed. For a sensitivity analysis we defined four properties as non-biomedical: diplomatic relation (P530), shares border with (P47), member of (P463) and country (P17). The first two come from the original paper; the last two were added by us. Together they make up 25.0% of the $\text{PMI} < 2$ set. This definition was made after inspecting the data, so every main result is reported with and without it.

4.4 Does PMI predict the verdict?

On the $\text{PMI} < 2$ set we computed Spearman's ρ between PMI and the binary outcome; the area under the ROC curve (AUC) of PMI as a score for true; logistic regressions of true on PMI, crude and with property fixed effects, with standard errors clustered by subject entity (the same subject recurs many times), reporting the odds ratio (OR) per +1 PMI; a within-property pooled AUC (property-level AUCs weighted by the number of positive–negative pairs) with a 95% interval from 500 stratified bootstrap replicates; and the property-adjusted OR for $\text{PMI} < 0$ versus $0 \leq \text{PMI} < 2$. Robustness checks removed exact duplicate triples and kept one copy of each unordered subject–object pair.

4.5 No association versus weak association

We fitted a logistic model of true on a no-association indicator plus property effects, restricted to biomedical properties, and compared the two groups property by property for properties with at least 100 rows in each group.

4.6 Internal consistency

A repeated identical prompt, or a symmetric relation, should receive one verdict. We counted duplicate triples and reciprocal diplomatic-relation pairs ($A \rightarrow B$ and $B \rightarrow A$) that received different verdicts.

4.7 Design of a threshold test

Because the supplied data end at PMI 2, the threshold itself cannot be tested. We specify a test for relations within a bandwidth of 2, with $c = \text{PMI} - 2$:

$$\text{logit } \Pr(\text{true}) = \beta_0 + \beta_1 c + \beta_2 \cdot 1[c \geq 0] + \beta_3 \cdot c \cdot 1[c \geq 0] + \text{property effects} \quad (2)$$

The coefficient β_2 is the jump in log-odds at the threshold, allowing a different slope on each side. The notebook implements this as `evaluate_threshold()`. We also give the sample size for a two-proportion comparison at 80% power and $\alpha = 0.05$, which assumes independent observations and therefore understates the real requirement.

4.8 Software and status of the analysis

The analysis used Python 3.12.3 with pandas 3.0.2, NumPy 2.4.4, SciPy 1.17.1, statsmodels 0.15.0 (Seabold and Perktold, 2010; Virtanen et al., 2020), matplotlib 3.10.8 and openpyxl 3.1.5; the random seed was 20241. The analysis was not pre-registered. Model choices and the non-biomedical definition followed inspection of the data, so the study is exploratory.

5. Results

5.1 Overall verdicts (RQ1, RQ2)

Table 3. Jev verdicts for the two flagged groups. Values are counts (percentage); 95% Wilson interval for true.

Group	n	true	maybe	false
PMI < 2	15,925	10,407 (65.3; 64.6–66.1)	163 (1.0)	5,355 (33.6)
No PubMed association	24,800	17,328 (69.9; 69.3–70.4)	196 (0.8)	7,276 (29.3)

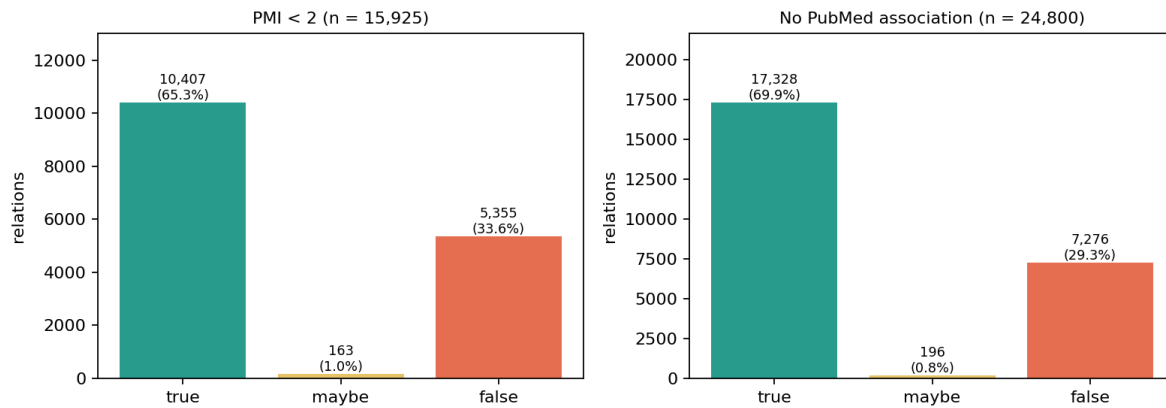


Figure 1. Distribution of Jev verdicts in the two flagged groups.

If Jev's verdicts are taken as the reference, about one third of the flagged relations are judged inaccurate (33.6% and 29.3%). The remainder are judged accurate or uncertain.

5.2 Accuracy rate per PMI value (RQ3)

Table 4. Share of relations judged true, by PMI value (rounded down). Values are % (95% Wilson interval).

PMI bin	All: n	All: % true	Biomedical only: n	Biomedical only: % true
[−6, −5)	4	25.0 (4.6–69.9)	4	25.0 (4.6–69.9)
[−5, −4)	25	36.0 (20.2–55.5)	25	36.0 (20.2–55.5)
[−4, −3)	162	41.4 (34.1–49.1)	137	31.4 (24.2–39.6)
[−3, −2)	579	57.5 (53.4–61.5)	386	44.3 (39.4–49.3)
[−2, −1)	1,543	63.7 (61.3–66.1)	881	45.9 (42.6–49.2)
[−1, 0)	3,041	66.6 (64.9–68.2)	1,947	52.1 (49.9–54.3)
[0, 1)	4,646	65.3 (64.0–66.7)	3,574	56.8 (55.2–58.4)
[1, 2)	5,925	66.7 (65.5–67.9)	4,990	61.9 (60.5–63.2)
−10 (no association)	24,800	69.9 (69.3–70.4)	23,812	69.2

Biomedical only excludes P530, P47, P463 and P17. The bins below PMI −4 hold only 4 and 25 relations.

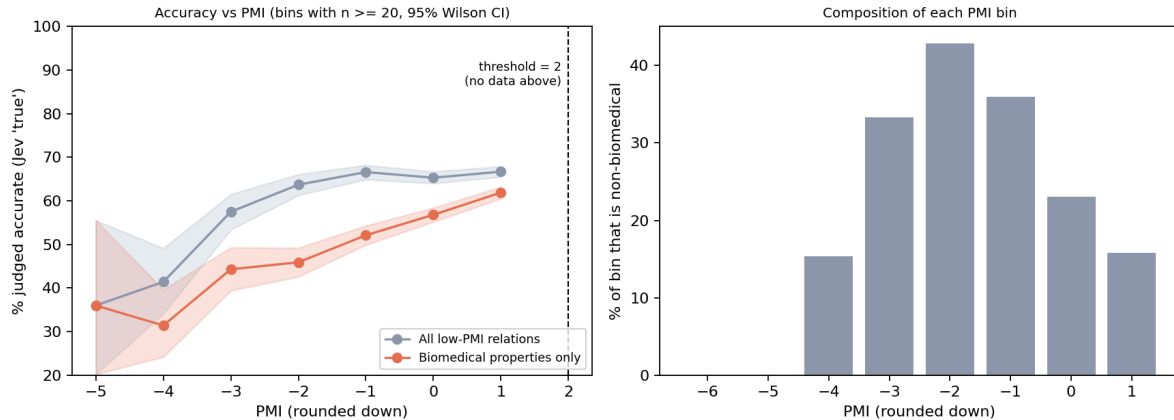


Figure 2. Left: accuracy rate against PMI, all relations and biomedical properties only (bins with $n \geq 20$; shaded bands are 95% intervals). Right: share of each PMI bin made up of non-biomedical properties.

Pooled over all relations the rate rises from about 41% at PMI -4 to 64% at -2 and then stays between 65% and 67% up to PMI 2. With non-biomedical properties removed it rises steadily to 61.9% in the bin just below the threshold (Figure 2).

The two curves differ because of composition. Non-biomedical properties, which Jev judges accurate far more often (diplomatic relation: 93.1%), make up 15.4%, 33.3%, 42.9%, 36%, 23.1% and 15.8% of the bins from $[-4, -3)$ to $[1, 2)$, peaking around PMI -2 . At moderate PMI the pooled rate is therefore inflated by a different mix of relation types.

5.3 Does PMI predict the verdict? (RQ4)

Table 5. Association between PMI and a true verdict (PMI < 2 group, $n = 15,925$). OR = odds ratio per +1 PMI unless stated.

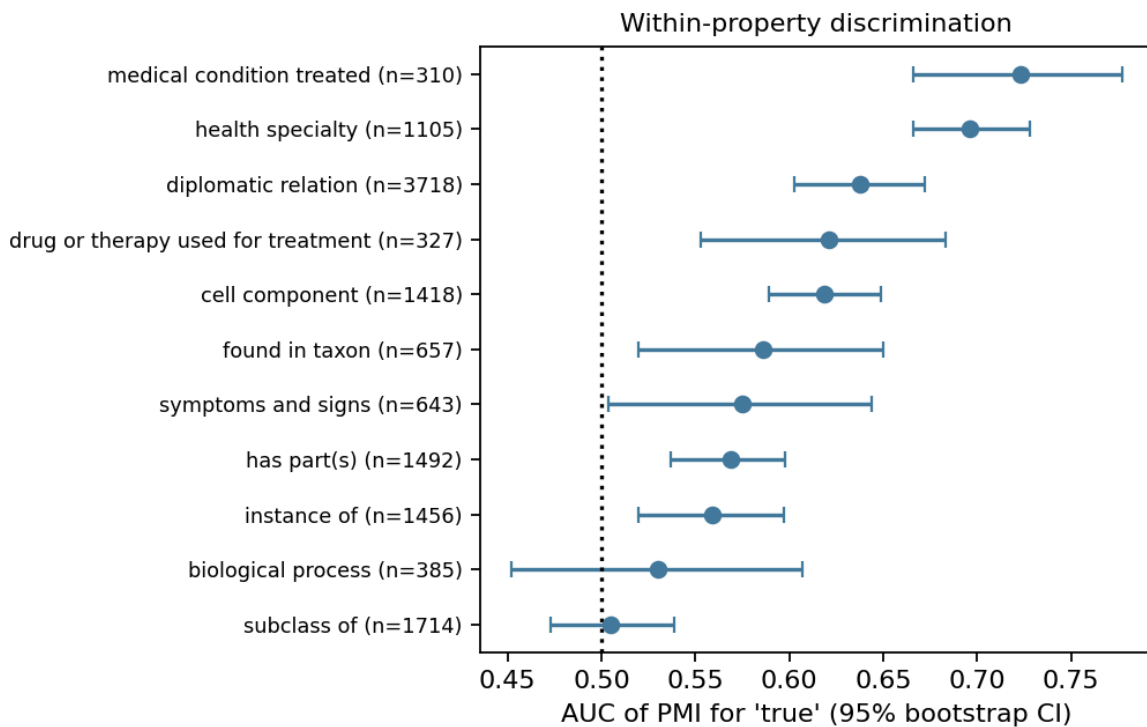
Measure	Value
Spearman ρ	0.03 ($p = 5.7 \times 10^{-5}$)
AUC, all relations	0.52
AUC, biomedical properties only	0.57
Within-property pooled AUC	0.60 (0.58–0.61)
OR, crude	1.08 (1.03–1.14)
OR, property-adjusted	1.37 (1.32–1.42)
OR, biomedical only, crude	1.23 (1.19–1.27)
OR, biomedical only, property-adjusted	1.35 (1.30–1.40)
OR for PMI < 0 vs. $0 \leq \text{PMI} < 2$, property-adjusted	0.50 (0.45–0.55)
Robustness: duplicates removed ($n = 15,879$)	1.37 (1.31–1.42)
Robustness: one copy per unordered pair ($n = 13,888$)	1.36 (1.31–1.41)

Intervals are 95%. Regression standard errors are clustered by subject entity.

Within a relation type, higher PMI goes with a higher chance of a true verdict, and negative PMI roughly halves the odds. The estimate is stable across the robustness checks. Discrimination is modest: a within-property AUC of 0.60 means a randomly chosen accurate relation has a higher PMI than a randomly chosen inaccurate one about 60% of the time. Table 6 shows that this varies strongly by property.

Table 6. Within-property discrimination of PMI for a true verdict (properties with $n \geq 300$; 1,000 bootstrap replicates).

Property	n	% true	AUC (95% CI)
medical condition treated (P2175)	310	45.8	0.72 (0.67–0.78)
health specialty (P1995)	1,105	44.7	0.70 (0.67–0.73)
diplomatic relation (P530)	3,718	93.1	0.64 (0.60–0.67)
drug or therapy used for treatment (P2176)	327	40.4	0.62 (0.55–0.68)
cell component (P681)	1,418	42.0	0.62 (0.59–0.65)
found in taxon (P703)	657	11.7	0.59 (0.52–0.65)
symptoms and signs (P780)	643	85.4	0.57 (0.50–0.64)
has part(s) (P527)	1,492	67.4	0.57 (0.54–0.60)
instance of (P31)	1,456	81.9	0.56 (0.52–0.60)
biological process (P682)	385	15.8	0.53 (0.45–0.61)
subclass of (P279)	1,714	76.6	0.51 (0.47–0.54)

**Figure 3.** AUC of PMI for a true verdict within each property, with 95% bootstrap intervals. The dotted line marks 0.5 (no discrimination).

PMI discriminates best for disease-oriented relations such as *medical condition treated* and *health specialty*, the kind of statement that PubMed co-indexing records. It is close to uninformative for *subclass of* and *biological process*, whose verdicts do not depend on how often two keywords are indexed together.

5.4 No association versus weak association (RQ5)

Among biomedical properties, relations with no PubMed association were judged accurate more often than relations with weak association: 69.2% ($n = 23,812$) versus 56.6% ($n = 11,944$). The

property-adjusted OR is **1.61** (95% CI 1.51–1.73). In 11 of the 14 properties with at least 100 rows in each group, the no-association group has the higher rate (Table 7, Figure 4).

Table 7. Share judged true, by property, for the two flagged groups (properties with $n \geq 100$ in each group).

Property	PMI < 2: n	% true	No assoc.: n	% true
health specialty (P1995)	1,105	44.7	1,608	53.5
medical condition treated (P2175)	310	45.8	702	80.1
drug or therapy used for treatment (P2176)	327	40.4	740	69.3
subclass of (P279)	1,714	76.6	4,687	75.0
subject has role (P2868)	238	42.9	5,034	75.7
instance of (P31)	1,456	81.9	2,830	93.9
has use (P366)	106	66.0	440	77.3
has part(s) (P527)	1,492	67.4	995	76.9
molecular function (P680)	227	57.3	144	68.8
cell component (P681)	1,418	42.0	1,200	42.5
biological process (P682)	385	15.8	496	33.1
found in taxon (P703)	657	11.7	2,065	37.9
significant drug interaction (P769)	178	91.0	687	79.3
symptoms and signs (P780)	643	85.4	510	84.9

Non-biomedical properties are excluded.

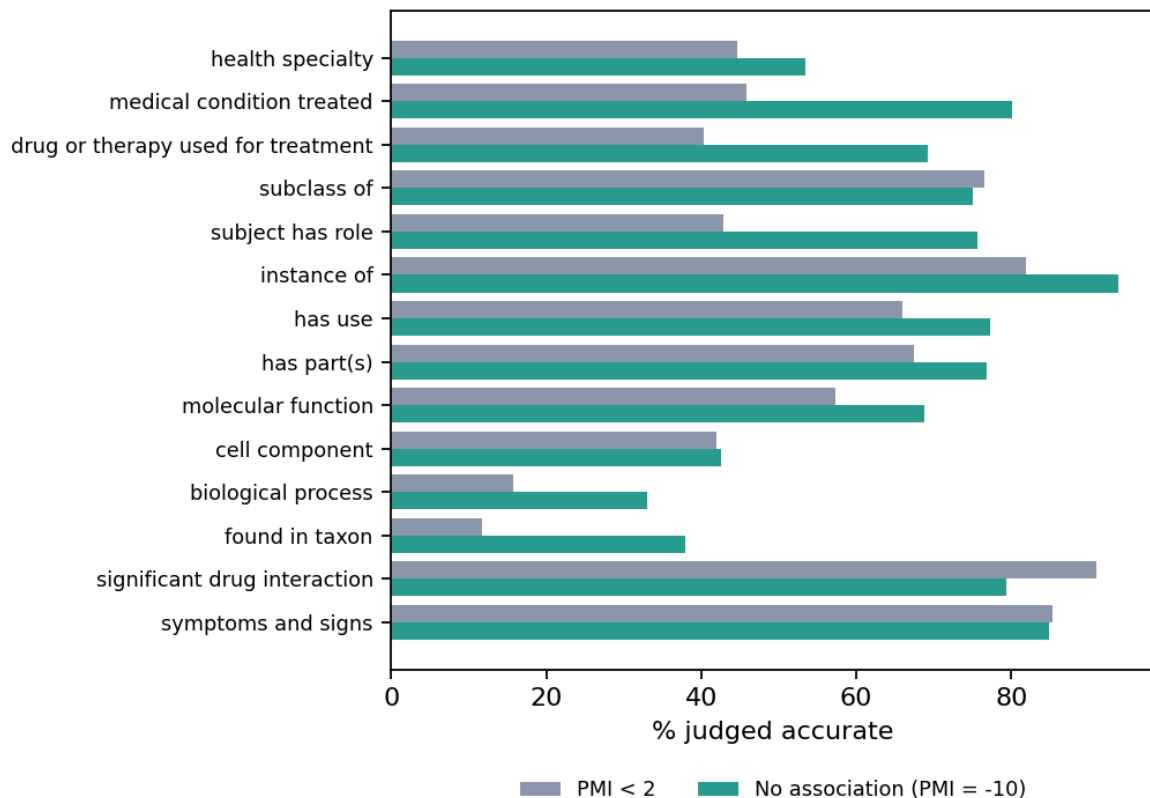


Figure 4. Share judged true for the same property in the weak-association (PMI < 2) and no-association (PMI = -10) groups.

5.5 Internal consistency

Table 8 counts verdicts on repeated or reciprocal inputs. Agreement is high, but a symmetric relation or a repeated identical prompt should never receive two verdicts. The disagreements show that verdicts are not fully reproducible even on identical input. Each relation was judged only once, so the size of this effect across all rows is unknown.

Table 8. Consistency of verdicts on repeated or reciprocal inputs.

Check	PMI < 2	No association
Duplicate triple groups	46	127
... with differing verdicts	7	4
Reciprocal diplomatic-relation pairs	1,836	445
... with differing verdicts	18	22

5.6 Sample size for testing the threshold

Starting from the 61.9% accuracy rate in the highest available biomedical bin, a two-sided test at $\alpha = 0.05$ with 80% power needs the numbers of labelled relations per side of PMI 2 shown in Table 9, assuming independent observations. The original study's figures imply roughly 55,600 relations at or above PMI 2 (96,341 – 40,725; our derivation, not a number stated in the paper), so labelling the required sample is feasible.

Table 9. Relations needed on each side of PMI 2 to detect a given difference in accuracy rate.

Difference to detect (percentage points)	Relations per side
2	9,157
3	4,046
5	1,438
10	346

6. Discussion

What the flagged sets contain. Under this judge, most flagged relations are judged accurate: 65% for weak association and 70% for no association. The original wording that relations below the threshold are “not considered valid” is stronger than these data support; its other statement, that they are flagged for human validation, fits them well. A reviewer working through the flagged list should expect roughly one relation in three to be judged inaccurate by Jev, not most of them.

PMI carries a real signal that is easy to lose. Pooled across relation types, PMI looks almost useless (unadjusted OR 1.08; AUC 0.52). Within relation type, the odds of a true verdict rise by about 37% per PMI unit and halve below zero. Relation type is a much stronger predictor of the verdict than PMI: the share judged accurate ranges from 12% to 93% across the larger properties. Any evaluation of PMI-based flags should therefore report results by relation type or adjust for it.

The threshold of 2 is not tested by these data. The curve for biomedical properties is still rising in the last bin before the threshold, with no visible plateau or jump, and there are no data above it. This neither supports nor refutes 2 as a boundary. A cut-off chosen as the mode of a distribution has no particular reason to coincide with a change in correctness, and the borrowed benchmarks concern word-pair relatedness, not the truth of knowledge-graph statements. A direct test needs verdicts above and below the threshold (Section 4.7).

“No association” is not “weak association”. The original paper groups both as deficient, but the data show they behave differently: no-association relations are judged accurate more often, within the same property. A plausible reason is that $N(x,y) = 0$ arises mainly from rarity. Of the no-association subjects, 71.5% are supplementary concepts, indexed in few articles, so a zero joint count says little about whether the relation is true. We could not test this directly because article counts per entity were not supplied.

Why a relation may be judged accurate despite low or no PMI. (i) The entities are very frequent or very rare in PubMed. (ii) The relation is taxonomic, compositional or geopolitical and is rarely the topic of an article. (iii) The Wikidata item maps only coarsely to the MeSH descriptor. (iv) The relation is established knowledge that needs no article-level support. (v) The judge may be lenient toward plausible-sounding triples.

What the results do not show. They do not show that any particular relation is true or false, that Jev is accurate, or that the results generalise beyond these sets. They do not show a difference in correctness between relations above and below PMI 2.

7. Limitations

1. **The outcome is not ground truth.** Jev saw three labels and no evidence. Relations between well-known entities may be judged accurate because the entities are familiar, and familiar pairs also tend to have higher PMI. Part of the PMI gradient may therefore reflect familiarity rather than correctness; the supplied data cannot separate the two.
2. **No human validation.** The precision and recall of the verdicts are unknown, so treating false as an error is an assumption.
3. **Single run, non-deterministic verdicts.** Table 8 shows identical or reciprocal inputs receiving different verdicts. Repeated runs would be needed to size the effect.
4. **Range restriction.** Only $\text{PMI} < 2$ and the -10 default are present, so no claim about $\text{PMI} \geq 2$ is supported.
5. **Post-hoc choices.** The non-biomedical property set and the model specification were chosen after inspecting the data. The headline estimates were stable with and without the non-biomedical set, but the analysis is exploratory.
6. **Entity resolution.** Labels are used without disambiguation, and some are bare identifiers. Chemical names include salt and stereoisomer variants. MeSH descriptors, supplementary concepts and qualifiers are treated alike.
7. **Non-independence.** Duplicates, reciprocal pairs and recurring subjects violate independence. Subject-clustered errors and the robustness checks address this in part, not fully. The sample-size calculation is therefore a lower bound.
8. **Sparse bins and timing.** Bins below PMI -4 hold 4 to 162 relations. Wikidata (snapshot of August 2023 in the original study), PubMed and the judging service reflect different dates.
9. **Upstream steps not repeated.** The PMI computation, MeSH mapping, label retrieval and verdict run were not repeated here; the analysis starts from the supplied files.

8. Conclusion

We evaluated the PMI flag of Turki et al. (2024) using one independent verdict for each of 40,725 flagged Wikidata relations. Most flagged relations (65.3% with $\text{PMI} < 2$; 69.9% with no PubMed association) were judged accurate, so the flag identifies a review queue rather than a list of errors. Within relation types, PMI predicts the verdict modestly but consistently (OR 1.37 per PMI unit;

within-property AUC 0.60), and this signal is hidden when relation types are pooled. Relations with no PubMed co-occurrence were judged accurate more often than weakly co-occurring ones, which argues against treating them as the same kind of deficiency.

Data and code availability

All numbers, tables and figures in this paper are produced by the accompanying Jupyter notebook (*Wikidata_PMI_Jev_Analysis.ipynb*), which reads the two verdict spreadsheets exactly as originally supplied (*Low_PMI.xlsx* and *Not_Available_in_PubMed.xlsx*). Outputs are written to a *results* folder. The notebook also contains, switched off by default, code to recompute PMI from Entrez counts, to fetch Wikidata labels and to re-run the verdicts; the Entrez query syntax there is our assumption because the original paper does not print it. The original study's source data and code are cited by its authors at <https://doi.org/10.6084/m9.figshare.24438184>.

Declarations

Use of AI. This paper and the notebook were drafted with Claude Sonnet 5.5 as an AI assistant; the authors are responsible for checking all content.

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